Olympiad Diophantine equations

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November 2026

L4-NT2 Level 4 Olympiad · Number theory

Proof methods for integer equations, namely modular obstructions, p-adic valuations, size arguments, descent and exponential equations.
By the end you can choose a modulus that rules out solutions and justify the choice completely; use valuations, Legendre's formula and squeezing between powers to bound solutions; write full BMO-standard solutions to exponential and descent problems.
Prerequisites: Modular arithmetic (L3-NT1), Integer equations (L3-NT2), Orders, Fermat and Euler (L4-NT1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Show that $x^2+y^2=3z^2$ has no solutions in positive integers. Try a few values and nothing works, but trying values proves nothing. Instead, look modulo 3. Squares are 0 or 1, so $x^2+y^2\equiv 0 \pmod 3$ forces $x^2\equiv y^2\equiv 0$, that is, $3\mid x$ and $3\mid y$. Then $9\mid 3z^2$, so $3\mid z^2$ and $3\mid z$. Dividing everything by 3 gives a new solution $(x/3,y/3,z/3)$ with a smaller z. If there were any positive solution, there would be one with z as small as possible, and we have just produced a smaller one. Contradiction.

The proof used a modulus in which squares are restricted, followed by descent. Most olympiad Diophantine equations are solved by combining a few standard methods, namely modular obstructions, p-adic valuations, size arguments, descent and factorisation. This unit covers each of them, and how to decide which to try first on a given equation.

1.1 Modular obstructions

A modulus is useful when the powers in the equation take very few values. The ones to know by heart:

- squares take the values $\{0,1\}$ modulo 3 and 4, $\{0,1,4\}$ modulo 8, and $\{0,\pm1\}$ modulo 5.
- cubes take the values $\{0,\pm1\}$ modulo 7 and modulo 9.
- fourth powers take the values $\{0,1\}$ modulo 5 and modulo 16.

These moduli come from Fermat's Little Theorem. If $k\mid p-1$ and $p\nmid x$, then $(x^k)^{(p-1)/k}=x^{p-1}\equiv 1$, so x^k is a root of $y^{(p-1)/k}\equiv 1$, which by Lagrange's Theorem has at most $(p-1)/k$ roots. So k-th powers take at most $1+\frac{p-1}{k}$ values modulo p. Against cubes, use 7 or 13. Against fifth powers, use 11. Against sixth powers, 7 or 13. Powers of a fixed base, such as 2^n or 3^n, are periodic with period equal to the order, so 3^n takes only the values $1,3,9$ modulo 13.

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Example 1.
Find all positive integers n and m with $n! + 3 = m^2$.

Solution.
Try small cases. Here $n = 1$ gives $4 = 2^2$, $n = 2$ gives 5, $n = 3$ gives $9 = 3^2$, $n = 4$ gives 27, $n = 5$ gives 123, and $n = 6$ gives 723, which lies between $26^2 = 676$ and $27^2 = 729$. So we expect only $n = 1$ and $n = 3$.
For large n, $n!$ is divisible by every small number, so modulo any fixed M the left side is eventually just 3. We therefore want a modulus in which 3 is not a square. Modulo 3 is the obvious first try, but modulo 3 fails, since $3 \equiv 0$ is a square. Modulo 4 works, because 3 is not a square modulo 4, and $4 \mid n!$ for all $n \geq 4$. So for $n \geq 4$ we would have $m^2 \equiv 3 \pmod 4$, which is impossible.
The solutions are $(n, m) = (1, 2)$ and $(3, 3)$. In general, a factorial or a high power is 0 modulo many numbers, so choose a modulus in which the remaining terms cannot be a square.

1.2 p-adic valuations

For a prime p and a nonzero integer n, $v_p(n)$ is the exponent of p in n. The rules are $v_p(ab) = v_p(a) + v_p(b)$ and $v_p(a + b) \geq \min(v_p(a), v_p(b))$, with equality when $v_p(a) \neq v_p(b)$ (factor out the smaller power of p, and one term is left divisible by p and the other is not). A positive integer is a perfect k-th power if and only if k divides every one of its valuations.

Coprime products.
If a, b are coprime positive integers and ab is a perfect k-th power, then a and b are both perfect k-th powers.
Legendre's Formula. $v_p(n!) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p-1}$, where $s_p(n)$ is the sum of the digits of n in base p.

Proofs. For coprime products, each prime divides at most one of a, b, so $v_p(a)$ is either 0 or $v_p(ab)$, a multiple of k in both cases. For Legendre, $v_p(n!) = \sum_{m \leq n} v_p(m)$, and $v_p(m)$ counts the $i \geq 1$ with $p^i \mid m$. Swapping the order of counting, for each i there are $\lfloor n/p^i \rfloor$ multiples of p^i up to n. For the digit form, if $n = \sum_j d_j p^j$ then $\lfloor n/p^i \rfloor = \sum_{j \geq i} d_j p^{j-i}$, and summing over $i \geq 1$ gives $\sum_j d_j \frac{p^j - 1}{p-1} = \frac{n - s_p(n)}{p-1}$. □

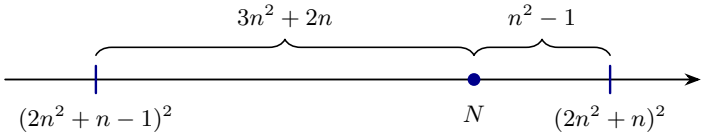
Example 2.
For which integers $n \geq 2$ is $n!$ divisible by 2^{n-2}?

Solution.
The direct approach is to estimate $\lfloor n/2 \rfloor + \lfloor n/4 \rfloor + \cdots$, which is a little less than n, and then control the error terms from the floors. The base 2 form of Legendre's Formula gives the error exactly, as $v_2(n!) = n - s_2(n)$, where $s_2(n)$ is the number of 1s in the binary expansion of n.
So we need $n - s_2(n) \geq n - 2$, that is, $s_2(n) \leq 2$. The answer is that n is a power of 2, or a sum of two distinct powers of 2. As a check, $3 = 2 + 1$ works since $2 \mid 6$, while $7 = 111_2$ fails because $v_2(7!) = 4 < 5$.

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1.3 Size and bounding

If $k^2 < N < (k+1)^2$ for some integer k, then N is not a square, so to show that an expression N is never a square, trap the expression between two consecutive squares. To find the right squares, take the square root of the leading terms. For $N = 4n^4 + 4n^3 + 1$, the top terms suggest $\sqrt{N} \approx 2n^2 + n$, and indeed $(2n^2+n)^2 = 4n^4 + 4n^3 + n^2$. Comparing, $(2n^2+n)^2 - N = n^2 - 1$ and $N - (2n^2+n-1)^2 = 3n^2 + 2n$, both positive for $n \geq 2$. So among positive integers, N is a square only for $n = 1$, when N equals 9.



$N = 4n^4 + 4n^3 + 1$: for $n \geq 2$ both gaps are positive, so no square fits.

Other size arguments to keep in mind are to order the variables ($x \leq y \leq z$) and see which term dominates, to use the fact that factorials and exponentials outgrow polynomials, and, in a factorised equation, each factor divides a fixed number, which leaves finitely many cases.

Example 3.
Find all positive integers x and y with $x^2 + 3^x = y^2$.

Solution.
With x in an exponent one might start reducing modulo small numbers, but $y^2 - x^2$ factorises as $(y-x)(y+x) = 3^x$. Since the right side is positive and $y + x > 0$, we have $y > x$, so both factors are positive, and hence both are powers of 3. Write $y - x = 3^i$ and $y + x = 3^{x-i}$, where $i < x - i$ because $y - x < y + x$.
Subtracting, $2x = 3^{x-i} - 3^i = 3^i(3^{x-2i} - 1)$. Put $j = x - 2i \geq 1$, so that $x = 2i + j$ and

$$2(2i + j) = 3^i(3^j - 1).$$

The left side is linear in i and j and the right side is exponential, so only small values can work.
We use the following inequalities, each true at its first value and preserved at every step, because the left side at least triples while the right side only grows by a constant. These are $3^j - 1 \geq 2j$ for $j \geq 1$ (and likewise $3^j - 1 \geq 4j$ for $j \geq 2$), and $3^i \geq 4i + 1$ for $i \geq 2$. If $i \geq 2$, the right side is at least $(4i+1) \cdot 2j = 8ij + 2j > 4i + 2j$, so there are no solutions. If $i = 1$, we need $4 + 2j = 3(3^j - 1)$. Here $j = 1$ works ($6 = 6$), and for $j \geq 2$ we have $3^j - 1 \geq 4j$, so the right side is at least $12j > 2j + 4$. If $i = 0$, we need $2j = 3^j - 1$. Here $j = 1$ works and $j \geq 2$ gives $3^j - 1 \geq 4j > 2j$.
The case $(i, j) = (0, 1)$ gives $x = 1$, $y = 2$, and $(i, j) = (1, 1)$ gives $x = 3$, $y = 6$. Both check, as $1 + 3 = 4$ and $9 + 27 = 36$. So the solutions are $(1, 2)$ and $(3, 6)$.

1.4 Infinite descent

Descent is induction in reverse. If every solution produces a smaller positive solution, there are no solutions at all, because positive integers cannot decrease for ever. Write the argument as a minimal counterexample, "take a solution with $|z|$ as small as possible", and then produce a smaller one. The usual setting is a homogeneous equation in which some prime is forced to

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divide every variable. For example, in $x^3 + 2y^3 = 4z^3$, x must be even, so writing $x = 2x'$ and dividing by 2 gives $4x'^3 + y^3 = 2z^3$, so y is even, and then z is even, and $(x/2, y/2, z/2)$ is again a solution. So the only solution is $x = y = z = 0$.

1.5 Exponential equations

For equations such as $3^a - 2^b = 1$ there are three standard approaches.

1. Factorise, typically as a difference of squares when an exponent is even, and control the gcd of the factors.
2. Reduce modulo a power of one base to determine the other exponent modulo an order.
3. Count the powers of a prime on each side, by parity arguments or LTE.

Example 4.
Find all non-negative integers a and b with $3^b = 5 \cdot 2^a + 1$.

Solution.
Small cases show that $3^b - 1$ runs $0, 2, 8, 26, 80, 242, 728, \ldots$, and only $80 = 5 \cdot 16$ has the right shape, giving $(a, b) = (4, 4)$. Rewrite the equation as $3^b - 1 = 5 \cdot 2^a$. The case $b = 0$ is impossible, since then the left side is 0. Reducing modulo powers of 2 gives information about a, but the factor 5 gives information about b, so we use the factor 5 first.
The powers of 3 modulo 5 are $3, 4, 2, 1$, so $\mathrm{ord}_5(3) = 4$, and by the Order Lemma of L4-NT1, $5 \mid 3^b - 1$ if and only if $4 \mid b$. Write $b = 4d$ with $d \geq 1$ and factorise as a difference of squares:

$$(9^d - 1)(9^d + 1) = 5 \cdot 2^a.$$

The two factors differ by 2 and are both even, so their gcd is 2 and the coprime products lemma cannot be applied as stated. Look at the larger factor on its own. Since $9 \equiv 1 \pmod 8$, we have $9^d + 1 \equiv 2 \pmod 8$, so $v_2(9^d + 1) = 1$. Also $9^d + 1$ divides $5 \cdot 2^a$, so its odd part divides 5. Hence $9^d + 1$ is 2 or 10. The first is impossible for $d \geq 1$, and the second gives $d = 1$.
Then $b = 4$ and $5 \cdot 2^a = 80$, so $a = 4$. The only solution is $(a, b) = (4, 4)$.

Common traps.

- "Integers" includes zero and negative numbers. Read the question and handle them.
- Finding solutions by testing is not a proof that there are no others. You must prove a bound or an obstruction.
- When you factorise, check the signs of the factors and their gcd before splitting into cases.
- The coprime products lemma needs coprimality. If the factors share a factor of 2, as in Example 4, deal with that factor of 2 first.
- A modular obstruction is only a proof if you show the complete list of residues.
- In descent, check that the new solution is smaller and still a genuine solution. Homogeneous equations always have the zero solution, which must be excluded or listed.

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1.6 How to spot this in a contest

Always test small cases first, since what you find decides the method. In a nutshell, I would proceed as follows.

1. If testing finds a few small solutions, the problem is about proving there are no large ones, so think factorisation (each factor divides something fixed) or squeezing between consecutive powers.
2. If testing finds nothing, look for a modulus, chosen by the powers present (4, 8, 3 or 5 for squares, 7 or 9 for cubes, and the order for fixed bases), or for descent if the equation is homogeneous.
3. Factorials suggest valuations and Legendre's Formula. A product equal to a power suggests valuations and the coprime products lemma.
4. Two exponential bases suggest working modulo a power of one base, then using orders, LTE or a difference of squares.

At BMO standard, a full solution states the answer, checks that the claimed solutions work, and then proves that there are no others. Markers look for every case to be closed, so make your case split explicit.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Prove that there are no integers x and y with $x^3 + y^3 = 2026$.

Problem 2 ⋆⋆ Find all positive integers n for which $3^n + 55$ is a perfect square.

Problem 3 ⋆⋆ When 2026! is written in base 12, how many zeros does 2026! end with?

Problem 4 ⋆⋆⋆ Prove that the only integer solution of $x^2 + y^2 + z^2 = 7w^2$ is $x = y = z = w = 0$.

Problem 5 ⋆⋆⋆ Find all positive integers n for which $n^4 + 2n^3 + 6n + 7$ is a perfect square.

Problem 6 ⋆⋆⋆⋆ Find all pairs of positive integers (a, b) such that $7^a - 3^b = 4$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1987 Problem 5** ⋆
   Rearrange so that one side factorises.
2. **UKMT BMO1 2001/02 Problem 1** ⋆
   Clear denominators and factorise, using the fact that n is odd.

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3. **UKMT BMO1 2016/17 Problem 3** ⋆⋆
   Multiply by 4, complete the square on each side, then factorise.

4. **USAMO 1979 Problem 1** ⋆⋆
   Fourth powers take very few values modulo 16.

5. **AIME 1989 Problem 9** ⋆⋆
   Pin n down with size estimates, then fix n with residues modulo small numbers.

6. **UKMT BMO1 2012/13 Problem 4** ⋆⋆
   Eliminate n between the two expressions to get a factorisable relation between the square roots.

7. **UKMT BMO1 2011/12 Problem 5** ⋆⋆⋆
   Pair up the factors, then trap one side between consecutive products.

8. **USAMO 1976 Problem 3** ⋆⋆⋆
   Work modulo 4, then descend.

9. **USAJMO 2011 Problem 1** ⋆⋆⋆
   Split on the parity of n and choose a modulus for each case.

10. **UKMT BMO1 2018/19 Problem 6** ⋆⋆⋆
    Find a closed form for the number of journeys, then compute its 2-adic and 5-adic valuations.

11. **USAJMO 2013 Problem 1** ⋆⋆⋆
    Cubes are very restricted modulo 9.

12. **UKMT BMO1 2006/07 Problem 6** ⋆⋆⋆⋆
    Name the square root and factorise. Coprime factors whose product is a square are squares.

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4 Hints

1. Cubes take very few values modulo 9 and modulo 7. Try both, since only one of them works.
2. First decide the parity of n by working modulo 4. Then factorise.
3. $12 = 2^2 \cdot 3$. Compute $v_2(2026!)$ and $v_3(2026!)$, and do not assume which prime is the bottleneck.
4. Squares modulo 8 are 0, 1 or 4. Split on the parity of w, and descend when w is even.
5. Compare with $(n^2+n)^2$ and $(n^2+n-1)^2$, and factorise the difference with the first.
6. If $b \geq 2$, work modulo 9 to find a modulo 3. Then look for a prime modulo which 3 has order 3.

5 Solutions

Problem 1

Prove that there are no integers x and y with $x^3 + y^3 = 2026$.

Modulo 9 does not help, since cubes are 0 or ± 1 modulo 9, so $x^3 + y^3$ can be any of $0, \pm 1, \pm 2$, and $2026 \equiv 1 \pmod 9$.

Work modulo 7 instead. If $7 \nmid x$ then $x^6 \equiv 1$ by Fermat's Little Theorem, so $x^3 \equiv \pm 1$, hence every cube is congruent to 0, 1 or 6 modulo 7. The possible values of $x^3 + y^3$ modulo 7 are therefore $0+0=0$, $0+1=1$, $0+6=6$, $1+1=2$, $1+6 \equiv 0$ and $6+6 \equiv 5$, that is, $0, 1, 2, 5, 6$. But $2026 = 7 \cdot 289 + 3 \equiv 3 \pmod 7$. So the equation has no integer solutions. □

Problem 2

Find all positive integers n for which $3^n + 55$ is a perfect square.

If n is odd then $3^n \equiv (-1)^n = -1 \pmod 4$, so $3^n + 55 \equiv 54 \equiv 2 \pmod 4$, which is not a square. So $n = 2k$ is even.

Write $3^{2k} + 55 = m^2$ with $m > 0$. Then $(m - 3^k)(m + 3^k) = 55$. The second factor is positive, so the first is positive too, and the first is the smaller of the two. The factorisations of 55 into two positive factors are $1 \cdot 55$ and $5 \cdot 11$. Subtracting the factors gives $2 \cdot 3^k = 54$ or $2 \cdot 3^k = 6$, so $k = 3$ or $k = 1$.

Thus $n = 6$, with $729 + 55 = 784 = 28^2$, and $n = 2$, with $9 + 55 = 64 = 8^2$. These are the only solutions.

Answer: $n = 2$ and $n = 6$

Problem 3

When 2026! is written in base 12, how many zeros does 2026! end with?

The number of trailing zeros in base 12 is the largest k with $12^k \mid 2026!$. Since $12^k = 2^{2k}3^k$, this is $k = \min\left(\lfloor v_2(2026!)/2 \rfloor,\ v_3(2026!)\right)$.

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In binary, $2026 = 1024 + 512 + 256 + 128 + 64 + 32 + 8 + 2 = 11111101010_2$, which has eight 1s. By Legendre's Formula, $v_2(2026!) = 2026 - 8 = 2018$. For the prime 3, $v_3(2026!) = 675 + 225 + 75 + 25 + 8 + 2 = 1010$.

So $k = \min(1009, 1010) = 1009$. Here the limiting factor is 4, not 3. By analogy with base 10, where 5 is always the limiting prime, one might assume the opposite.

Answer: 1009

Problem 4

Prove that the only integer solution of $x^2 + y^2 + z^2 = 7w^2$ is $x = y = z = w = 0$.

Suppose there is a solution other than the zero one. If $w = 0$ then $x^2 + y^2 + z^2 = 0$ forces $x = y = z = 0$, so in a nonzero solution $w \neq 0$. Choose a nonzero solution with $|w|$ as small as possible.

Recall that an odd square is 1 modulo 8 and an even square is 0 or 4 modulo 8. If w is odd, then $7w^2 \equiv 7 \pmod 8$. But a sum of three numbers from $\{0, 1, 4\}$ is one of $0, 1, 2, 3, 4, 5, 6, 8, 9, 12$, which are $0, 1, 2, 3, 4, 5, 6, 0, 1, 4$ modulo 8. This sum is never 7 modulo 8. This is a contradiction.

If w is even, then $7w^2 \equiv 0 \pmod 4$. Squares are 0 or 1 modulo 4, so $x^2 + y^2 + z^2$ is congruent modulo 4 to the number of odd terms among x, y, z, which is $0, 1, 2$ or 3. This number must be 0, so x, y, z are all even. Then $(x/2, y/2, z/2, w/2)$ is an integer solution with $0 < |w/2| < |w|$, contradicting the minimality of $|w|$.

Both cases are impossible, so the only solution is $x = y = z = w = 0$. □

Problem 5

Find all positive integers n for which $n^4 + 2n^3 + 6n + 7$ is a perfect square.

Let $f(n) = n^4 + 2n^3 + 6n + 7$. The leading terms suggest comparing with $(n^2 + n)^2 = n^4 + 2n^3 + n^2$, and

$$f(n) - (n^2 + n)^2 = -n^2 + 6n + 7 = (7 - n)(n + 1).$$

Also $(n^2 + n - 1)^2 = n^4 + 2n^3 - n^2 - 2n + 1$, so $f(n) - (n^2 + n - 1)^2 = n^2 + 8n + 6 > 0$.

For $n > 7$ the first difference is negative and the second positive, so $(n^2 + n - 1)^2 < f(n) < (n^2 + n)^2$, and $f(n)$ lies between consecutive squares, so $f(n)$ is not a square. Hence $n \leq 7$.

For $n = 7$ the first difference is 0, so $f(7) = 56^2 = 3136$ is a square. For $n = 1, \ldots, 6$ the values are $16, 51, 160, 415, 912, 1771$. The first is 4^2, and the others are not squares, since $49 < 51 < 64$, $144 < 160 < 169$, $400 < 415 < 441$, $900 < 912 < 961$ and $1764 < 1771 < 1849$. So the answer is $n = 1$ and $n = 7$.

Answer: $n = 1$ and $n = 7$

Problem 6

Find all pairs of positive integers (a, b) such that $7^a - 3^b = 4$.

The pair $(1, 1)$ works, since $7 - 3 = 4$. We show there are no others. If $b = 1$ then $7^a = 7$ and $a = 1$, so suppose $b \geq 2$.

Then $9 \mid 3^b$, so $7^a \equiv 4 \pmod 9$. The powers of 7 modulo 9 are $7, 4, 1$ and then repeat, since $7^3 = 343 \equiv 1$. Hence $a \equiv 2 \pmod 3$.

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Now we want a modulus in which 3^b takes few values and in which knowing a modulo 3 restricts 7^a. A prime p with $\operatorname{ord}_p(3) = 3$ must divide $3^3 - 1 = 26$, and $p = 13$ works, since $3^3 = 27 \equiv 1 \pmod{13}$, so $3^b \in \{1, 3, 9\}$ modulo 13. Modulo 13, $7^2 \equiv 10$, $7^3 \equiv 70 \equiv 5$ and $7^6 \equiv 25 \equiv -1$, so $\operatorname{ord}_{13}(7) = 12$, a multiple of 3. Since $a \equiv 2 \pmod{3}$, a is congruent to $2, 5, 8$ or 11 modulo 12, and

$$7^2 \equiv 10, \quad 7^5 = 7^3 \cdot 7^2 \equiv 50 \equiv 11, \quad 7^8 = 7^6 \cdot 7^2 \equiv -10 \equiv 3, \quad 7^{11} = 7^6 \cdot 7^5 \equiv -11 \equiv 2.$$

So $3^b = 7^a - 4$ is congruent to one of $6, 7, 12, 11$ modulo 13, none of which lies in $\{1, 3, 9\}$. This contradiction shows $b \geq 2$ is impossible.

Therefore the only solution is $(a, b) = (1, 1)$. □

Answer: $(a, b) = (1, 1)$

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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